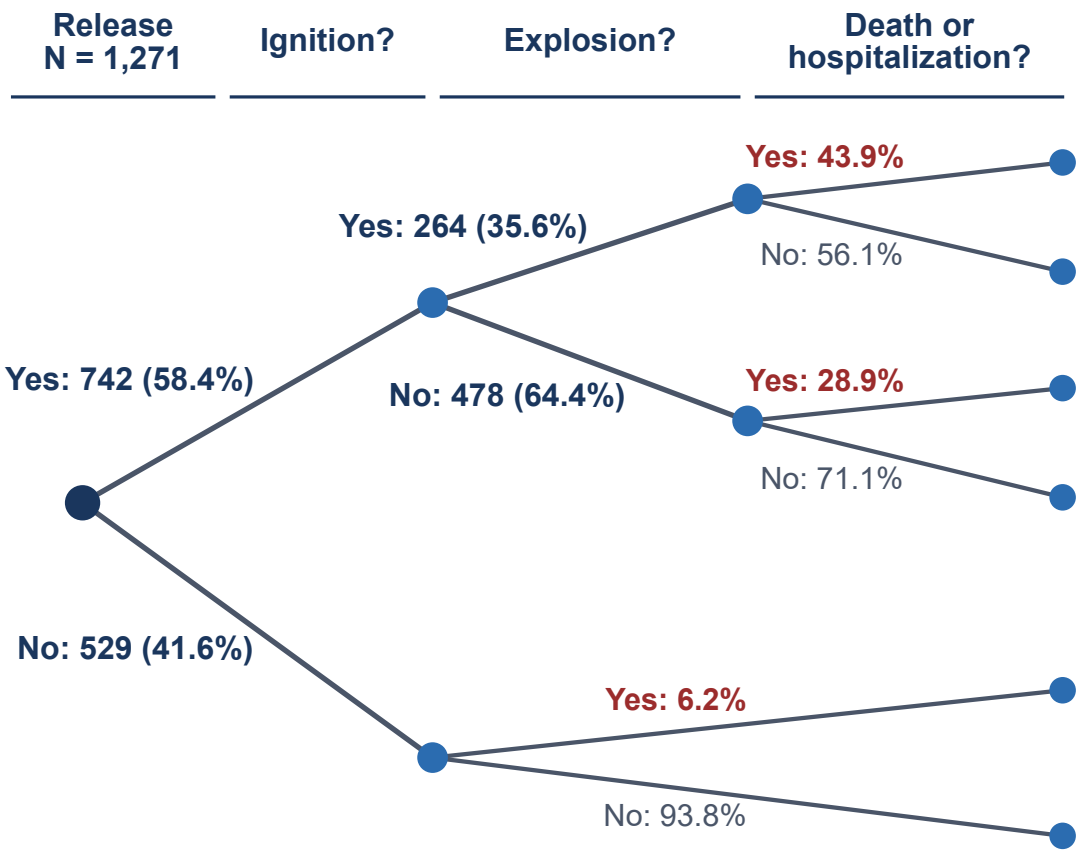
Event tree of reportable gas distribution incidents

PHMSA records, 2010–2024, N = 1,271



Outcome path: incidents (share), casualties

Path 1: fire and explosion, severe harm
116 (9.1%), 67 deaths, 308 hospitalized

Path 2: fire and explosion, no severe harm
148 (11.6%), 0 deaths, 0 hospitalized

Path 3: fire, no explosion, severe harm
138 (10.9%), 39 deaths, 193 hospitalized

Path 4: fire, no explosion, no severe harm
340 (26.8%), 0 deaths, 0 hospitalized

Path 5: no ignition, severe harm
33 (2.6%), 7 deaths, 36 hospitalized

Path 6: no ignition, no severe harm
496 (39.0%), 0 deaths, 0 hospitalized

Shares are of all 1,271 incidents; the six paths add up to 100%. All 113 deaths and 537 hospitalizations are on Paths 1, 3 and 5.